Experiment 13

Network Troubleshooting Using CLI Tools

Lab 13 Objectives:

● Use CLI tools (ping, tracert, nslookup) for network troubleshooting. Test connectivity, resolve domain names, and trace packet routes.

Prerequisites: The prerequisites of this lab are-

● Basic understanding of networking concepts (IP addressing, DNS, routing).
● Familiarity with command-line tools for network diagnostics.

Outcomes: By the end of this lab, students will be able to:

● Test network connectivity using ICMP-based tools.
● Resolve domain names to IP addresses and troubleshoot DNS issues.
● Trace packet routes to identify network path issues.

Introduction

Network troubleshooting is a critical process used to identify, diagnose, and resolve connectivity issues within computer networks. Efficient troubleshooting ensures that devices can communicate properly, data flows securely, and services remain available without interruptions. Command Line Interface (CLI) tools such as **ping**, **tracert/traceroute**, and **nslookup/dig** are fundamental utilities used by network administrators and technicians to perform this troubleshooting.

- **Ping** helps test whether a host is reachable and measures the time it takes for packets to travel back and forth, revealing latency and packet loss issues.
- **Tracert/traceroute** maps the path packets take from the source device to a destination, showing every network device (router) they pass through. This helps detect where delays or failures occur in the network path.

- **Nslookup/dig** queries DNS servers to resolve human-readable domain names (like google.com) into IP addresses, verifying that DNS services are functioning correctly.

By using these tools together, network professionals can quickly pinpoint problems related to connectivity, routing, and DNS resolution, which are common sources of network errors.

Experiment Description

1. Using ping

The **ping** command works by sending Internet Control Message Protocol (ICMP) echo request packets to a specified IP address or domain name. The target host responds with echo reply packets if it is reachable.

- **Purpose:**
  To verify network connectivity between the source device and a target host.
- **What it measures:**
    - **Round-trip time (RTT):** The time taken for a packet to reach the destination and return, usually measured in milliseconds (ms).
    - **Packet loss:** The percentage of packets sent that do not receive a reply, indicating potential network congestion or faults.
- **Use cases:**
    - Checking if a website or server is reachable.
    - Measuring network latency and performance.
    - Detecting packet loss that may cause slow or unreliable communication.
- **Example command:**
  ping google.com

2. Using tracert/traceroute

The **tracert** (Windows) or **traceroute** (Linux/macOS) command shows the path that packets take to reach a destination host. It works by sending packets with incrementally increasing Time-To-Live (TTL) values, forcing each intermediate router to respond with a "time exceeded" message.

- **Purpose:**
  To map the complete route from the source to the destination and measure the delay at each hop (router or gateway).
- **What it shows:**
  - The IP addresses or hostnames of all routers between source and destination.
  - The delay (latency) in milliseconds for each hop.
- **Use cases:**
  - Identifying where packets are delayed or lost in the network path.
  - Detecting network bottlenecks or points of failure.
  - Troubleshooting routing issues or misconfigurations.
- **Example command:**
  - Windows:
    tracert google.com
  - Linux/macOS:
    traceroute google.com

3. Using nslookup/dig

The **nslookup** or **dig** command queries Domain Name System (DNS) servers to translate domain names into IP addresses.

- **Purpose:**
  To verify DNS resolution and troubleshoot DNS-related problems.
- **What it provides:**
  - The IP address associated with a domain name.
  - Information about DNS servers and their responses.
- **Use cases:**
  - Checking if DNS servers are functioning correctly.
  - Verifying the IP address that a domain resolves to.
  - Diagnosing DNS misconfigurations or failures that prevent web access.
- **Example command:**
  nslookup google.com

4. Troubleshooting Steps Using These Tools

Ping:
Look for packet loss or high latency values. Packet loss indicates dropped packets, which can cause slow or unstable connections. High latency may suggest network congestion or long-distance communication delays.

Traceroute:
Analyze the route hops for any unreachable routers (marked by asterisks *) or significantly high response times. This can identify problematic segments of the network.

Nslookup:
If the domain name does not resolve to an IP address, it may indicate DNS server issues. Verify DNS settings and try alternative DNS servers if necessary.

Cross-location testing:
Run the same tests from different network locations or devices to determine if the problem is localized or widespread.

CLI Commands and Examples
Ping:
ping -c 4 google.com
Sends 4 ICMP echo requests.
Checks if the host responds and measures response times.

Traceroute (Linux/macOS):
traceroute google.com
Shows each hop's IP and latency from source to destination.

Tracert (Windows):
tracert google.com

Nslookup:
nslookup google.com
Queries the DNS server for the IP address of google.com.

Observations

- **Ping Results:**

    - Successful replies with low latency indicate good connectivity.
    - Packet loss or timeouts show network issues.
- **Traceroute Results:**
    - A normal route shows a series of hops with reasonable delays.
    - Timeouts or high delays at certain hops indicate potential network congestion or failures.
- **Nslookup Results:**
    - Correct IP address returned confirms DNS is working.
    - Failure or incorrect IP suggests DNS server issues or misconfigurations.
    -

Troubleshooting Solutions

- If **ping** shows packet loss, check physical connections, restart routers, or contact ISP.
- If **traceroute** shows timeouts at specific hops, those routers might be down or blocking traceroute packets.
- If **nslookup** fails, verify DNS server settings or switch to a public DNS server (e.g., Google DNS 8.8.8.8).
- Use **ipconfig** (Windows) or **ifconfig** (Linux/macOS) to check local IP configurations.